Finding Buyers: Information and Trade Promotion in the First Globalization

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Abstract

Governments spend significant resources on export promotion policies, yet RCTs and non-experimental studies disagree on whether they move firms into new markets. Furthermore, since modern programs bundle many services, it is difficult to pin down which components work. I isolate one: information about potential foreign buyers. During the first globalization, the Austro-Hungarian Export Compass published consular lists of foreign buyers that rotated across destination markets, generating variation in when exporters could identify buyers in each destination. Using a novel panel of 13,000 firms from 1890 to 1898, I find that buyer information raises the probability of exporting to a covered destination the following year by 5.9%. Since I observe what information exporters received from buyers, I uncover what information on buyers may be used: effects are stronger for nearby markets and persist beyond one year only if listings contain a larger share of intermediary traders.

Keywords: buyer-supplier links, export promotion, search frictions, industrial policy, economic history

JEL Codes: F14, N73, O14, L25

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1 Introduction

Governments allocate significant resources to public and private export promotion agencies, motivated by the direct and indirect benefits of exporting. Whether these agencies actually help firms enter foreign markets remains debated. Beyond the general difficulty of evaluating export promotion policies, a further challenge is disentangling the effects of different support channels. I focus on an information channel operating through fixed costs: learning about potential foreign buyers.

I use a historical quasi-experiment from late nineteenth-century Austria-Hungary to study how access to buyer lists affects firms' export decisions and through which channels the effect operates. Since I observe the exact information export-ready firms received, it is possible to conduct a variety of heterogeneity analyses to study what kind of buyer-information helped firms form new market connections, or make their business relations persist.

As Habsburg commercial interests expanded beyond the well-connected markets, Austria-Hungary invested heavily in trade-facilitating institutions from the 1870s onward. Consular offices became a key component on trade facilitation, and from 1890 were explicitly tasked to collect information on local importing firms, on top of providing general market intelligence. The Ministry of Commerce consolidated the results into a widely available registry — the Export Compass — published annually from 1890 to 1898, then biennially until 1915.

The setting generates quasi-experimental variation in buyer information availability. Each year, consular offices in a rotating set of destination markets were instructed to compile lists of locally active importers — recording their name, address, imported products, type (wholesalers or retailers), and any Austro-Hungarian affiliation. Crucially, the rotation schedule was determined administratively by the Ministry of Commerce rather than in response to observed trade flows, making coverage of a given destination in a given year plausibly exogenous to firm-level export potential. This staggered, rotating coverage across seven — rather arbitrary — regions (from the Balkans to South America) allows me to estimate changes in export activity within Austro-Hungarian firms, using firm-year, destination-year, and firm-destination fixed effects to absorb time-varying demand conditions and persistent bilateral trade relationships.

To measure the effect of buyer information on export decisions, I construct a firm-year-destination country panel using the editions of the *Export Compass* from 1890 to 1898, covering 13,190 firms. A firm f receives a signal[1] in year t if the *Export Compass* lists at least one buyer in destination d that imports a product matching firm f's exports — for instance, a wine producer receives a signal from Romania in a year when Romanian wine importers appear in the registry. The outcome variable is export status in year $t+1$: whether firm f exports to destination d following the signal.

Results show a significant increase in export status for destination country d at time $t+1$

[1]Throughout this paper, the term *signal* refers to the arrival of buyer information from a destination country in a given year, as published in the *Export Compass*. This differs from the standard use of the term in information economics, where signals are strategic actions taken by informed parties to communicate private information (see e.g. Spence, 2002). Here, the signal is an exogenous information shock received by the exporter from a third-party institution.

conditional on receiving a signal from d at time t. On average, there is a 0.289 percentage point increase (from including none to using all fixed effects) compared to a baseline probability of 0.049, which corresponds to a 5.9% increase in the probability of a firm exporting to d at $t+1$. As expected, stronger signals—defined as a larger number of potential buyers—correspond to an increase in probability of up to 12% (in the saturated model) when the number of potential buyers exceeds 60, while weaker signals, especially with buyer numbers below 20, have smaller magnitude. The treatment shows no pre-trends or anticipatory effects: current signals do not predict current export status, and future signals do not predict future export status. The only significant result comes with a year delay, which is consistent with the information publication of the Export Compass: even if a firm exported the year it received a signal, their report on export destinations would only appear in subsequent editions of the Export Compass.

Interestingly, the effect does not persist—at $t+2$ and $t+3$ it is no longer statistically significant. Two candidate mechanisms could explain this decay. First, if buyer information becomes publicly available over time, firms that did not receive a signal may subsequently enter the market, eroding the treated firm's advantage. Second, firms may fulfill short-term contracts satisfying local demand before reallocating to more lucrative destinations, or leave export markets if they are not profitable (Albornoz et al., 2023). These mechanisms have distinct implications: the first suggests that the signal generates social value beyond its private recipient, while the second points to strategic destination-switching by firms. To distinguish between them, further analysis of firm-level behavior will be conducted.

The effect of buyer information depends on where the buyers are and who they are. Across destination markets, the response is concentrated where information is novel and cheapest to act on: signals from Mediterranean destinations, readily served from the ports of Trieste and Fiume, roughly double the baseline effect, while point estimates for distant overseas markets are close to zero—consistent with information alone being insufficient to overcome the higher fixed costs of serving remote destinations.

The data allows me to further decompose the information signals firms receive based on the Austro-Hungarian nationality of the buyer. Ex ante, such buyers should be more valuable counterparties for two reasons: first, cultural and linguistic differences can increase trade frictions (Guiso et al., 2009; Melitz, 2008; Melitz and Toubal, 2014), since business negotiation in a foreign language is more difficult and different cultural norms may negatively impact business agreements and transactions. Second, contract enforcement becomes more difficult if domestic exporters are unaware of the legal and regulatory environment of foreign markets, and lack basic information on the creditworthiness of buyers (Anderson and Marcouiller, 2002; Nunn, 2007). This is especially relevant in countries with less stable institutions (Levchenko, 2007), and ethnic networks have been shown to reduce precisely these frictions (Rauch and Trindade, 2002). At the same time, information about compatriot buyers is likely to be less novel, since exporters could learn about such firms through domestic networks even without the registry. In the data, these forces appear to offset: I find

no robust evidence that signals with a high share of Austro-Hungarian buyers were differentially effective.

Besides the Austro-Hungarian nationality of potential buyers, the occupation (or business type) of foreign importers is extensively recorded, distinguishing merchant intermediaries from direct buyers. The type of importer matters for what the information signal represents: a listed merchant buyer opens a different entry pathway than a direct buyer, and may be more likely to aggregate demand across multiple domestic suppliers (Ahn et al., 2011). This distinction turns out to be quantitatively central: signals dominated by merchant intermediaries—buyers purchasing wholesale or explicitly listed as traders—raise the probability of export entry roughly four times more than other signals, and the pattern is robust to alternative definitions of merchant dominance. The buyers most worth knowing about were thus not final customers but intermediaries, consistent with merchants serving as entry points that aggregate demand and lower the cost of forming a first foreign relationship. Furthermore, having a high share of merchants within a signal results in a longer-lasting presence in foreign markets: even signals containing 50% of more merchants two or three years before still predict a higher export presence a year later.

Finally, I aggregate the analysis to the firm level to ask whether buyer information expanded firms' export portfolios or merely redirected them. Two patterns emerge. First, the geographic scope of exporting responds to the depth of information rather than its breadth: the total number of matched buyers a firm could identify across markets raises its number of export destinations in the following year, while the mere count of signaled destinations does not. Because total destination counts rise with buyer information, the destination-level entry effects do not reflect pure reallocation across markets—the Compass expanded exporting on net rather than shifting it from uncovered to covered destinations. Second, firms receiving signals from more destinations narrow their exported product range, consistent with specialization: identified foreign demand allowed firms to concentrate on the product lines for which buyers were known to exist. Together, the firm-level results suggest that public buyer information changed not only where firms sold, but also what they sold.

This project contributes to three strands of literature. First, export promotion policies have been widely evaluated across countries (Volpe Martincus and Carballo, 2010; Volpe Martincus et al., 2010; Benguria, 2021; Broocks and Van Biesebroeck, 2017; Buus et al., 2025; Van Biesebroeck et al., 2015; Hiller, 2012; Görg et al., 2008; Schminke and Van Biesebroeck, 2013), with non-experimental studies—typically based on matching or difference-in-differences designs—finding mostly positive effects on both the intensive and extensive margins of exporting as well as on firm outcomes (with Bernard and Jensen (2004) a notable exception). Randomized evaluations paint a less optimistic picture: Breinlich et al. (2017); Kim et al. (2018); Ali et al. (2025); González et al. (2025) find no effect on the extensive margin, with at most modest effects on the intensive margin. Two explanations could reconcile the discrepancy: unobserved confounders inflating the non-experimental estimates, or interventions in the experimental studies that are too weak or too short-lived to move entry decisions. The treatments studied are consistent with the latter: Breinlich et al. (2017),

Cusolito et al. (2023), and Kim et al. (2018) evaluate one-off provision of general information about export practices rather than sustained, actionable intelligence about specific counterparties, and Ali et al. (2025), who study a substantial subsidy covering up to 50% of fixed entry costs such as trade-fair participation and foreign representation, find effects only on exports to already-established destinations. My setting speaks directly to this divide. The Export Compass combined features that neither side of the literature observes together: quasi-experimental variation from the rotation schedule, which addresses the confounding concern in observational studies, and a treatment that was specific, actionable, and repeated—annual lists identifying individual buyers by name, location, and product scope, published at national scale for nearly a decade. The positive extensive-margin effects I find suggest that the null results in recent experiments may reflect the nature of the information provided rather than the irrelevance of information: generic export training does not tell a firm whom to sell to; a buyer list does.

A concurrent paper, Kim and Kim (2026), studies KOTRA's public provision of foreign buyer information in South Korea (1977–1990) at the destination-industry level, finding positive export effects rationalized by a search-and-matching model. My firm-level setting is complementary: I observe individual firms' entry decisions and the content of each buyer listing, and I recover a comparable result in a very different historical and institutional context. Furthermore, my research question considers the puzzle regarding the extensive margin of export, as opposed to the intensive margin results in Kim and Kim (2026).

Within export promotion, the trade effects of diplomatic institutions such as embassies and consulates have been studied at the aggregate level, typically in gravity frameworks (Rose, 2007; Moons and van Bergeijk, 2017). These studies establish that diplomatic presence correlates with bilateral trade but cannot observe what diplomats actually do for firms. My contribution to this literature is to open that mechanism at the micro level: the consular service's information-gathering work is observed directly—in the form of the buyer lists it produced—and linked to the export decisions of individual firms, showing that a specific, low-cost output of diplomatic infrastructure moved firms into new markets.

Second, a growing literature documents that search and matching frictions between buyers and sellers are a first-order impediment to trade, comparable in magnitude to physical transport costs (Bernard and Moxnes, 2018; Allen, 2014; Lenoir et al., 2023; Eaton et al., 2022), with buyer and seller heterogeneity shaping matching outcomes and the density of potential counter-parties governing how quickly new matches form (Bernard et al., 2018; Miyauchi, 2024). Causal evidence confirms that reducing these frictions has economically significant consequences: the elimination of information lags raises trade volumes and reduces price dispersion (Steinwender, 2018); experimental reductions in search costs increase trade flows and producer revenues (Bergquist et al., 2024); and improved buyer–seller matching raises product quality and firm profits (Atkin et al., 2017; Cai et al., 2024). A common limitation of this literature, however, is that information is treated as binary—either available or not—and decompositions of frictions into distinct components (search,

contracting, cultural proximity) remain rare (Startz, 2024). This paper addresses that gap directly: exploiting a historical buyer registry with staggered, rotating destination coverage, I observe not merely whether buyer information was disclosed, but what it disclosed—buyer identity, product scope, intermediary status, and cultural proximity—allowing us to identify which dimensions of counter-party information are most consequential for market entry.

Third, evidence on industrial policies, including policies reducing information frictions in the 19th century (such as Steinwender (2018)), are rare, due to both the lack of micro-data on firms' export activity and the scarcity of natural experiments that empowers researchers to identify the effectivity and efficiency of policy tools. The analysis of industrial policies are especially difficult not only due to endogeneity issues, but also due to the fact that policies are never implemented randomly by real policy makers (Juhász et al., 2024). Reduced form evidence on trade protection shows a consistent positive effect of technology adoption (Juhász, 2018) and firm outcomes (Hanlon, 2020), however, these papers use semi-random variation resulting from exogenous events rather than from the implementation of policy tools. This project, similar to Bekes et al. (2026), analyses a policy aimed at assisting firms fostering business connections by reducing information frictions on several domain. While Bekes et al. (2026) analyses a large-scale world fair in Paris in 1900, the policy in this project has a different approach: information is provided each year from different angles of the known world, thus firms have the opportunity to learn about the markets and potential buyers through several years, choosing an export strategy that fits their needs and competencies best. To my best knowledge, there are no other papers analyzing export promotion (or information services) that use 19th century micro data. However, other mechanisms are explored using well-identified settings, such as trademark protection in Alfaro et al. (2022), intellectual property protection in Moser (2005, 2012), domestic market integration in Donaldson and Hornbeck (2016); Donaldson (2018), or the effects of colonialism in Romero (2023).

The paper is structured as follows. Section 2 describes the institutional context of trade in the first globalization: falling trade barriers, the transportation and communication revolutions, and the state-provided information services—above all the consular network—that emerged to reduce search and matching frictions in international markets. Section 3 presents the data sources and the construction of the firm-level panel, including the LLM-assisted parsing and product matching that assigns treatment, along with descriptive statistics on signals and export probabilities. Section 4 lays out the empirical strategy: the quasi-random rotation of destination coverage, the baseline specification, the confounders the design absorbs, and the remaining threats to identification. Section 5 presents the main results and reports robustness checks, including placebo signals, sample restrictions, and a continuous treatment measure. Section 6 traces out mechanisms through heterogeneity in signal strength, destination geography, and buyer composition. Section 7 aggregates the analysis to the firm level to examine effects on firms' overall export portfolios. Section 8 summarizes the results and connects them to policy discussions.

2 Trade institutions in the first globalization

By the end of the 19th century, several historical barriers to international trade had been substantially reduced: transport costs fell with the spread of steamships and railways (O'Rourke and Williamson, 1999), communication frictions declined with the telegraph (Steinwender, 2018), and postal infrastructure was internationalized through the Universal Postal Union established by the Treaty of Bern in 1874. Shipping times were reduced significantly (e.g. by 79% between Liverpool and New York (Hatton, 2025)), which also induced reduction in freight rates by up to 30% between 1870-1914 (Estevadeordal et al., 2003). International trade was increasing throughout the second half of the 19th century until WWI, before plummeting throughout the world wars (Klasing and Milionis, 2014).

Building on these technological and infrastructural innovations, trade institutions emerged facilitated by domestic commercial chambers and ministries, resulting in increased foreign representation for national business interests (Rose, 2007). In parallel, businessmen created new organizations for information sharing for network building, including export associations and private networking events (Eichenberger and Péry-Kaulmann, 2022). In addition, trade fairs and museums of commerce helped businesses finding trustworthy buyers and suppliers.

Due to the increasing demand for information on foreign markets, collecting information on foreign markets became widespread across industrialized countries. One of the most important institutions for increasing trade between countries and collect market intelligence was consular offices. As opposed to embassies, one country had multiple consular offices, usually presented in important commercial ports and trading hubs (de Goey, 2014). By the turn of the twentieth century, several hundred such offices operated across all continents [2]. Consuls collected information about tariffs, foreign market conditions, exchange rates and currencies, foreign trade associations, postal and telegraph fees, transportation costs including shipping rates and freight companies, and on domestic economic conditions in general. The Export Compass used in this project is an example of information collection used to foster trade relations.

According to Malfatti di Monte Tretto (1904), beyond these periodic reports, consuls were obligated to file ad hoc special reports on a defined set of topics including railway developments, public tenders and contract announcements, fraudulent firms, tariff changes, and bankruptcy notices. Crucially, consuls were required to respond to commercial inquiries about foreign buyers, with standardized instructions specifying the elements of a creditworthiness assessment, the choice of references, the required format, and the procedural distinction between Austrian and Hungarian enquirers reflecting the dual-monarchy's constitutional structure.

The institutional pipeline through which consular intelligence reached exporters is also specified in Malfatti di Monte Tretto (1904). Periodic trade reports were formally routed from consular offices through the Ministry of Foreign Affairs to the two trade ministries, and subsequently published

[2] Appendix Figure B.1 shows the European network for illustration

via the k.k. Handelsmuseum in Vienna, the Hungarian periodical *Consuli Közlemények*, and the *österreichisch-ungarische Konsularkorrespondenz* (Malfatti di Monte Tretto, 1904, p. 209). The training of future consular officials was itself organized around commercial competence: consular attachés were systematically assigned to Chambers of Commerce in major trading cities—including Vienna, Budapest, Trieste, Fiume, Prague, Brünn, and Reichenberg—where a designated chamber expert was responsible for their commercial and trade-policy education. This institutional architecture confirms that the Export Compass and the *Berichte der k.u.k. Konsularämter* were not ad hoc publications but downstream products of a mandated, hierarchically organized intelligence system, in which the identification and assessment of foreign buyers was a formal and regulated duty of the consular service.

The information flow specific for the buyer lists is shown in Figure 1. Consular offices collected information on local importers, which, through the institutional pipeline were synthesized in the Export Compass. Firms in Austria-Hungary had the opportunity to access and book and learn about the importers, and identify potential targets for their sales. In case the address and contact details were not indicated in the book (which often was the case), firm could reach out directly to the consular offices, as these foreign representatives had an obligation to supply information to firms upon request. Once firms received additional information directly from the consular offices, they could contact foreign importers though agents, telegraph, or post.

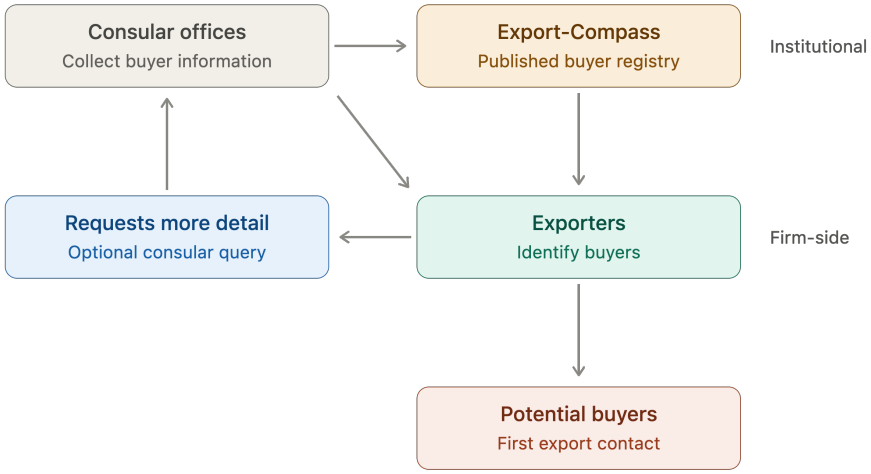


Figure 1: Information flow between firms and institutions

3 Constructing a historical firm-level panel

The empirical analysis builds on a newly constructed dataset from the *Austro–Hungarian Export-Compass* (*Export-Compass*), a government-adjacent trade publication produced by export-promotion institutions in cooperation with chambers of commerce and the consular network. An early version of the directory, limited to Viennese firms, appeared in 1888. By 1890, the publication expanded to cover firms throughout Austria–Hungary and organized them by industry and chamber jurisdiction. The 1890 volume also marks the first systematic inclusion of information on foreign importing firms (potential buyers).

The *Export-Compass* served two complementary purposes. First, it helped foreign buyers identify suppliers in Austria–Hungary by providing an organized directory of exporting firms and their product scope. Second, it supplied Austro–Hungarian firms with market intelligence intended to facilitate entry into, and continued operation in, foreign markets. Beyond firm directories, the publication compiled practical information for international commerce, including tariffs, exchange rates, measurement systems, consular locations, and contact information for export associations and museums of commerce.

A central component of the publication was its buyer-information service. Each year, different consular offices were instructed to collect or update lists of importing firms in a specific set of destinations, including the firms' location and the product groups they imported. Coverage rotated across world regions over time. This paper focuses on the volumes from 1890 to 1898: 1890 is the first year with comprehensive coverage of exporters across Austria–Hungary, and by 1895 the buyer lists span all major regions covered by the publication cycle. From 1896 onward, the reporting sequence restarted with the same region covered in 1890, consistent with a rotating coverage design.

3.1 Domestic exporting firms

Between 1890 and 1898, exporting firms are observed annually, yielding nine cross-sections that can be linked into a firm-level panel. The data contains 13,190 unique firms, of which 7,545 report at least one export destination at least once during the period. For each firm-year, the directory records the firm name and address, a description of exported products, and the destination countries served in that year. In some cases, entries also include additional details such as exhibition participation, advertised specialties, or secondary business locations.

3.2 Foreign importing firms (potential buyers)

The buyer lists cover importing firms located abroad and are not restricted to importers of Austro–Hungarian goods. Instead, they document firms with ongoing import activity in the destination market. For each importer, the directory reports the firm's location and the products or product categories imported; in some cases, it also records the nationality of the owner. If an importing

Table 1: Firm-Year Destination Coverage and Category Counts by Year

Year	Firms	Cov. rate	Avg. dest.	Dest. countries	Categories
1890	2,960	82.4%	3.0	58	413
1891	5,019	78.5%	2.8	59	469
1892	5,423	48.6%	2.6	59	488
1893	6,652	39.3%	2.5	59	518
1894	7,534	34.1%	2.4	55	533
1895	8,190	32.9%	2.4	55	534
1896	8,719	33.3%	2.4	57	545
1897	9,190	31.7%	2.5	57	544
1898	8,510	32.4%	2.4	56	542

Notes: *Firms* is the number of distinct firms listed in that edition; counts are of distinct firms and therefore exclude duplicate listing entries. *Cov. rate* is the share of those firms naming at least one export destination. *Avg. dest.* is the mean number of distinct destination countries named. *Dest. countries* is the number of distinct destination countries named by any firm that year. *Categories* is the number of distinct product categories assigned that year under the cleaned German category vocabulary. The decline in the coverage rate after 1891 reflects how completely destinations were recorded across editions rather than firms ceasing to export. Section **??** reports robustness to excluding individual editions.

firm had an affiliation with Austria–Hungary—for example, through Austro–Hungarian protection or a formal connection to the monarchy—this was explicitly noted in the registry. In addition, the market activity of the importing firm is indicated: for example, *detail* firms sold their products to consumers, while *en gross* firms purchased goods in large quantities and supplied smaller local firms. *Agents* and *commissioners* often acted as representatives or merchants of larger organizations.

3.3 Entity resolution across years

One challenge is to find firms across datasets, as there is no identification number available to match firms. To create a panel, the names of firms are used to match them across years. Since data collection was relatively consistent, many firms are successfully matched between years. The dataset covers between 3,173 and 10,320 firm-year observations per edition after removing within-year duplicates (see Table D.1).To track firms over time, names are matched across all year-pairs (e.g. 1890-91 or 1893-96) using exact and fuzzy string matching. Of the 151,708 total cross-year matches identified, 87.6% are exact and the remainder are fuzzy with a mean match score of 99.36—indicating that most name discrepancies reflect minor spelling variations rather than substantive mismatches. Match quality remains consistently high for consecutive year pairs, with mean scores all exceeding 99.6 (see Table D.2). Firm persistence varies considerably across the panel. Nearly 30% of firms appear in only a single edition, pointing to high turnover in the exporter population, while 8.1% are observed in all nine editions and form a stable core of persistent exporters.

Table 2: Descriptive Statistics: Import Demand Signals by Year

	N		Share of firms (%)	
Year	Firms	Products	Merchants	Austro-Hung.
1890	8,586	157	6.5	6.5
1891	4,489	186	8.6	2.2
1892	1,827	125	20.2	4.5
1893	3,592	173	12.1	0.9
1894	3,793	208	14.4	2.8
1895	1,607	128	21.3	1.7
1896	10,116	249	6.9	4.0
1897	4,027	191	5.3	2.3
Total	38,037	339	9.3	3.7

Notes: more information on which countries were covered in these years are found in Table 4.

3.4 Treatment and outcomes

Treatment is defined using firms' observed export product mix in a given year. On the demand side, each destination's consular list of importing firms is coded into roughly 550 product categories. Exporting firms are then matched to destinations whenever at least one importer in that destination is recorded as demanding the firm's exported product(s). For instance, the Bulgarian lists contain 17 importing firms classified as buyers of candles in 1890; consequently, every Austro-Hungarian candle exporter in that year receives a "signal" indicating potential buyers in Bulgaria. Because firms may export multiple products, a single firm can receive multiple signals—either from the same destination across different products or from different destinations altogether.

This procedure imitates the information-gathering process available to contemporary firms: having purchased the Export Compass, a firm would read through the relevant consular lists and identify potential buyers. The LLM matching replicates this process by linking firms' exported products to importers' demanded products [3]. Table 3 reports the annual number of signals and their prevalence in the firm–year–destination panel (i.e., the share of firm–year–country cells with at least one signal). Signal counts vary over time since each years covered a different region and number of countries, and the density of consular network also varied across countries. Further information about data cleaning can be found in Appendix C.

[3]See an example in Appendix Figure B.5

Table 3: Yearly summary statistics at the firm–destination–year level

	Panel size		Share of observations (%)			Firms
Year	Obs.	Firms	Signal_t	Export_t	Export_{t+1}	signaled (%)
1890	63,920	1,598	10.4	7.4	6.9	87.6
1891	112,280	2,807	7.7	7.2	5.5	88.3
1892	111,480	2,787	6.0	6.8	4.9	80.4
1893	120,720	3,018	3.0	6.2	4.6	83.8
1894	124,200	3,105	6.8	5.8	4.6	86.2
1895	131,760	3,294	3.7	5.7	4.6	67.4
1896	140,160	3,504	10.4	6.0	4.7	88.6
1897	142,280	3,557	4.6	6.0	4.7	84.5
Total	946,800	—	6.3	6.3	4.9	83.0

Notes: The table is constructed at the firm–destination–year level. Obs. counts firm–destination–year observations and Firms counts unique firms observed in that year; the firm count is not additive across years, so no total is reported. The three share columns give the percentage of firm–destination–year observations with $\text{signal}_t = 1$, $\text{export}_t = 1$ and $\text{export}_{t+1} = 1$ respectively. Firms signalled gives the percentage of firms observed in that year with at least one signalled destination, and is therefore a share of firms rather than of observations. The Total row pools all 946,800 observations from 1890 to 1897; the corresponding figure for firms signalled pools firm–year observations. 1898 is observed for export status but lies outside the signal data window and is excluded.

4 Empirical Strategy

4.1 The quasi-random variation

The empirical strategy exploits quasi-experimental design in the availability of buyer information generated by the annual publication cycle of the *Export-Compass*. Through its global consular network, Austria–Hungary collected and published directories of importing firms (potential foreign buyers). Importantly, the registry did not cover all destinations every year: instead, information provision rotated across world regions in a staggered manner, so that in a given year only a subset of destinations was covered. This institutional feature creates year-to-year variation in which destination markets become "information-rich" for exporters. There are two main reasons to believe that listing buyers is exogenous to export potential: first, using regions, instead of countries strongly indicates a methodological decision, as countries covered in each year are unlikely to have a correlated demand shock. Even through countries were more-or-less from the same part of the world, they still had a diverse political and economic systems, and macroeconomic shocks were likely only weakly correlated between countries, thus a boom in British-India were not likely to strongly correlate with China and Japan. Second, the rotating nature of the region coverages also suggest a simple methodological choice instead of careful consideration of economic dynamics or lobbying. As shown in Table 4, in 1896, the same countries are covered as in 1890, 1897 covers the countries from 1891, albeit missing Spain, Portugal, and Malta. 1898 covers the countries from 1892, except replacing Hongkong with Korea. Since there was not publication in 1899, and the publication of destination countries were ceased in 1900, the last country covered is 1898.

Table 4: Countries covered

Year	Countries covered
1890	Egypt, Bulgaria, Greece, East-Rumelia*, Serbia, Romania, Turkey
1891	Italy, Spain, Portugal, Algeria, Morocco, Tunisia, Malta*
1892	Indies, Hongkong*, China, Japan, Australia
1893	Canada, USA
1894	Switzerland, Russia, Sweden, Norway
1895	Chile, Peru*, Paraguay*, Argentina, Honduras*, Cuba*, Uruguay*, Brazil
1896	Egypt, Bulgaria, Greece, East-Rumelia*, Serbia, Romania, Turkey
1897	Italy, Morocco, Algeria, Tunisia
1898	Indies, China, Japan, Korea, Australia

Notes: Countries marked by * are covered in the Export Compass, however, due to the negligible export to these countries, they are not included in the analysis. Countries in 1898 are also not included, since there was no publication of Export Compass in 1899, the next and last version including export destination countries is in 1900.

I interpret the introduction of buyer listings for a destination country as an information shock that lowers search and matching frictions for market entry. Firms are differentially exposed to this

shock because the information is product-specific: a firm is treated in destination d and year t if the registry contains buyer information relevant to the firm's product scope for that destination. As explained in detail above, the main identifying variation therefore comes from changes over time in the set of covered destinations, interacted with firm-level product relevance.

4.2 Baseline Specification

I estimate how the provision of buyer information in year t affects the probability that firm f enters destination d in year $t+1$ using a panel at the firm–destination–year level:

$$\text{Export}_{fd,t+1} = \beta \, \text{Signal}_{fdt} + \gamma_{dt} + \alpha_{ft} + \phi_{fd} + \varepsilon_{fdt}, \tag{1}$$

where $\text{Export}_{fd,t+1}$ is an indicator for export status for firm f at destination country d at time $t+1$. Signal_{fdt} indicates whether the *Export-Compass* lists buyers relevant to firm f's product scope in destination d in year t. Destination-year fixed effects γ_{dt} absorb time-varying demand and trade-cost shocks in each market. Firm-year fixed effects α_{ft} capture changes in firms' overall export capacity and supply conditions. Firm–destination fixed effects ϕ_{fd} account for persistent bilateral relationships such as geography, networks, or long-run market fit. In some specifications, I additionally control for lagged export status Export_{fdt} to account for persistence in exporting. Standard errors are clustered at the firm level, however, alternative clustering choices are shown with the main results.

Under the identifying assumption that the rotation of destination coverage is orthogonal to firm-specific export shocks conditional on the fixed effects, β recovers the causal effect of access to state-provided buyer information on subsequent export entry. This design addresses the two main confounds in observational trade data—systematic differences in firm export ability and heterogeneous destination-level shocks—by exploiting within-firm, within-destination variation in information availability over time.

4.3 What the design absorbs: key confounding factors

The empirical specification is designed to address the major confounding concerns that arise in observational studies of exporting.

First, destination markets experience time-varying demand and trade-cost shocks that simultaneously affect export outcomes and the perceived usefulness of foreign market intelligence. Destination-year fixed effects γ_{dt} absorb any shock common to all firms exporting to a given destination in a given year: macroeconomic booms or recessions (e.g., rising demand in Egypt in 1892), exchange-rate movements, political events, tariff changes, new shipping connections, or diplomatic and trade-promotion initiatives. The estimated effect is therefore not driven by treated destinations becoming more attractive for reasons unrelated to information provision.

Second, firms differ in export capacity over time due to changes in productivity, ownership, technology adoption, or scale. Firm-year fixed effects α_{ft} control for such time-varying firm-level factors, so identification does not rely on comparing structurally different firms (e.g., fast-growing firms) to others, but on within-firm reallocation of export activity across destinations.

Third, exporting is shaped by persistent bilateral factors specific to a firm–destination pair: geographic distance, language, established commercial networks, diaspora links, and long-run product–market fit. Firm–destination fixed effects ϕ_{fd} absorb these time-invariant determinants, so the coefficient on buyer information is identified from changes within the same firm–destination pair over time rather than from cross-sectional differences across pairs.

Finally, export participation is highly persistent: firms that exported to a destination in the past are more likely to continue due to sunk costs, learning, or established relationships. Lagged export status Export_{fdt} accounts for this path dependence and separates the effect of new information from mechanical continuation.

Taken together, the identifying variation is sharply delimited: the effect of buyer information is estimated by comparing firms whose products match the newly listed buyers to firms in the *same destination and year* without such a match, within the same firm–destination pair over time. Threats to identification must therefore operate at the product–destination or firm–destination level—narrower than in a standard destination-level design. The next subsection discusses these remaining threats.

4.4 Threats to identification and remaining concerns

Non-random coverage and selection of destinations. The primary concern is that coverage rotation was not exogenous: the compilers might have prioritized booming markets or destinations where Austro-Hungarian exports were already expanding. Given the design, selection of entire destinations cannot drive the estimates; the relevant threat is selection on *product-specific* prospects—for instance, if consular offices collected more buyers of glassware in Serbia precisely because glass manufacturers were preparing to enter that market. Two pieces of evidence speak against this. First, the rotation schedule was institutionally predetermined: the Compass followed a six-region cycle between 1890 and 1895, and the 1896 preface explicitly states the completion-then-restart rule governing re-coverage. Contemporary reviews described the rotating coverage as "eine Specialität des Export-Compass," a fixed editorial design rather than a response to current trade conditions. Second, if coverage tracked latent product-level demand growth, buyer counts from the *following* year's edition should predict entry today; Table A.2 shows they do not.

Reverse causality and lobbying. A related concern is that firms or trade associations requested buyer lists for particular destinations, so that coverage responded to firms' export plans. The institutional setting leaves little room for this channel. The buyer lists were compiled by k.u.k. consular offices abroad and released by the Ministry of Foreign Affairs to the publisher; individual

exporters had no role in selecting destinations, and the rotation schedule was set in advance for the full cycle. The publication timing reinforces this: entries were current to the end of July and published in October, so the information could affect export behavior only from the following year onward—which is why the outcome is measured at $t+1$—leaving no scope for coverage to respond to contemporaneous entry decisions.

Anticipation. Because the rotation schedule was announced, firms could in principle anticipate coverage of a destination and delay entry until the buyer lists appeared, or prepare entry in advance. Anticipatory delay would bias the estimates upward; advance preparation would generate effects before publication and bias them downward. The lead placebo in Table A.2 bounds both concerns: export behavior shows no response to information that has not yet been published.

Bundling with other trade-promotion measures. Destination-wide policy initiatives coinciding with publication are immaterial by the same logic as above; the residual concern is product- or firm-targeted intervention bundled with the Compass. Here the nature of the treatment matters: the Compass was an informational tool published privately by Alexander Dorn's Volkswirtschaftlicher Verlag, providing buyer identities, locations, product scopes, and commercial functions—no financial support, no logistical assistance, and no firm-specific services accompanied publication. Any effect of the listings therefore reflects information, not a bundled program.

Measurement error in treatment assignment. Assigning treatment requires mapping firm product scopes to listed buyers' product descriptions, which is noisy: product categories in the registry are sometimes broad, and descriptions incomplete. Misclassification of this kind attenuates the estimates toward zero, so the reported effects are conservative.

Spillovers and information diffusion. The published registry was a public good: firms could share the lists, and intermediaries could act on them. Such diffusion contaminates the control group—untreated firms partially receive the treatment—and biases the estimates toward zero. In this case, the coefficients should therefore be read as a lower bound on the full effect of information provision.

Sample selection and outcome measurement. The outcome is constructed from the Compass's own exporter directory, so the specific concern is whether the quality of *exporter-side* reporting co-moves with *buyer-side* coverage rotation—if destination coverage years coincided with more complete recording of export destinations, the estimates would partly reflect measurement. The exporter directory was compiled annually for the full set of firms, on a cycle independent of the buyer-side rotation.

5 Results

5.1 Main results

Table 5 shows the effect of receiving a signal at time t on export status at time $t+1$. Standard errors are clustered at the firm level; the columns add fixed effects progressively. Without any fixed effects (column 1), the signal coefficient is 2.89 percentage points. Adding country–year fixed effects (column 2) reduces it to 1.31, adding firm–country fixed effects (column 3) to 0.301, and the saturated specification with all three pairwise fixed effects (column 4) yields 0.289 percentage points, significant at the 1% level.[4] The decline across columns is expected: most of the raw correlation between signals and subsequent exporting reflects destination-year conditions and persistent firm–destination relationships, which the fixed effects progressively absorb. The sharp drop between columns (2) and (3) is informative in itself: once firm–destination pairs are held fixed, three quarters of the remaining correlation disappears, indicating that the bulk of the naive association is cross-sectional sorting of firms into markets they were already serving rather than time-varying destination conditions. The saturated estimate—identified solely from comparing firms with and without product-matched buyers within the same destination-year, within the same firm–destination pair over time—is the preferred specification. Relative to the baseline entry probability of 4.9 percent, it implies that receiving a signal raises the probability of exporting to the destination in the following year by roughly 5.9 percent.

Table 5: Effect of consular buyer information on next-year exporting

	(1)	(2)	(3)	(4)
	export$_{t+1}$			
signal$_t$	2.84***	1.19***	0.275**	0.287***
	(0.135)	(0.197)	(0.113)	(0.104)
	[0.105]	[0.150]	[0.0879]	[0.0842]
	{1.50}	{0.339}	{0.163}	{0.130}
Observations	946,800	946,800	934,240	934,240
R^2	0.001	0.113	0.795	0.819
Firm–country FE	No	No	Yes	Yes
Firm–year FE	No	No	No	Yes
Country–year FE	No	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in $t+1$, thus units are percentage points. Standard errors clustered at the firm level in parentheses; two-way clustered at the firm and destination–year level in square brackets; clustered at the destination level in braces, with wild-cluster-bootstrap p-values. Clusters: 7,545 firms, 320 destination–years, 40 destinations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

[4]The fixed effects absorb 72.6% of the variation in signal$_t$; the coefficient is identified from the remaining 27.4%, which corresponds to variation in signaling across destinations within a firm–year and across years within a firm–destination pair.

Standard errors are clustered at the firm level throughout. A firm chooses its export markets jointly rather than destination by destination, and unobserved firm characteristics—e.g. managerial capacity, credit access, or an established shipping agent—persist across the sample and affect every destination the firm faces, thus errors are correlated across all cells belonging to the same firm. Clustering on the firm accommodates this dependence in both directions while leaving 7,545 independent units, comfortably above the range in which cluster-robust variance estimators become unreliable. The correction matters: heteroskedasticity-robust standard errors that ignore within-firm dependence are roughly half as large, and would overstate the precision of the estimate by a factor of about two. Because the product composition of each edition's buyer list is common to all firms facing a given destination in a given year, the last two rows of each column also report standard errors that are two-way clustered at the firm and destination–year level and clustered at the destination level; the latter is a conservative bound with only 40 clusters, so its p-value comes from a wild cluster bootstrap. The estimate remains significant under all clutering specifications.

5.2 Placebo treatment

Table 6 tests whether the baseline effect of signal_t on export_{t+1} holds when using a placebo signal for receiving a counterfactual buyer information before the actual information arrives. Since both the list of exporters and importers come from the same source, it should not be possible for an exporter to list its destinations before learning about potential importers. Even if an exporter updates its export destination list, the updated version should appear in the Export Compass in the subsequent year.

The leads show no anticipatory effects — signal_t on current export and signal_{t+1} on future export, and future signal on past export are uniformly insignificant statistically. The small magnitude of the coefficient also suggest insignificant economic effect, not only a result from decreased sample size. These are clean falsification results supporting the importance of signals on export status a year later.

5.3 No effect for signals received two years or before

To analyze whether signals have longer-term effect, I regress the signals received one or two years before the baseline signal. For this reason, singleton firm–country cells (17,400 observations) are dropped. In Table 7, the sample is restricted to outcome years for which the relevant lag window is observed: 677,120 observations where signal_{t-1} is available and 545,240 where signal_{t-2} is also available. Columns (1) and (3) re-estimate the baseline specification on these restricted samples without lags, so that the effect of adding lagged signals is separated from the effect of the sample restriction itself. Lagged signals are set to zero for outcome years whose lag window precedes the start of the signal data, on the grounds that no signaling opportunity existed before 1890.

The effect of receiving relevant buyer signal is concentrated at a one-year horizon. In column

Table 6: Effect of buyer-information placebo signals on next-period exporting

	(1)	(2)	(3)	(4)
		export$_{t+1}$		
signal$_t$	0.248***	0.228**	0.289***	0.235*
	(0.0896)	(0.0951)	(0.110)	(0.127)
signal$_{t+1}$		−0.0980		−0.152
		(0.0970)		(0.127)
signal$_{t+2}$				−0.155
				(0.135)
Observations	723,520	723,520	510,400	510,400
R^2	0.912	0.912	0.909	0.909
Sample	t+1	t+1	t+2	t+2
Firm–country FE	Yes	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in t + 1, thus units are percentage points. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(2), the estimate is 0.42 percentage points, indicating that the firm–destination pairs observed over longer stretches of the panel respond more strongly to signals; the comparison of columns (1) and the baseline in Table 5 isolates this composition effect from any role of the lag structure. Within the restricted samples, adding lagged signals leaves the contemporaneous coefficient essentially unchanged—0.42 to 0.427 percentage points in column (2) and 0.435 to 0.467 in columns (3)–(4)—while the lagged terms themselves are small and statistically indistinguishable from zero: 0.032 percentage points for signal$_{t-1}$ in column (2), and 0.078 and 0.125 percentage points for signal$_{t-1}$ and signal$_{t-2}$ in column (4), against standard errors of 0.134 to 0.163. Taken together, the estimates indicate that a signal affects export entry in the year immediately following its receipt, with no detectable additional effect from signals received two or three years earlier.

5.4 Past export status and export market entry

Table 8 reports specifications that interact and restrict the sample by past export status to the specific destination country in the year the signal is received. Column (1) augments the main specification from the last column of Table 5 with an indicator for exporting to the destination in year t: the signal coefficient is essentially unchanged, neither statistically nor in magnitude. Including a lagged dependent variable in a fixed-effects panel is normally discouraged because of the well-known Nickell bias, but the concern is muted here, since export status in t is predetermined

Table 7: Persistence of signal effect beyond 1 year

	(1)	(2)	(3)	(4)
	export$_{t+1}$			
signal$_t$	0.420***	0.427***	0.435***	0.467***
	(0.120)	(0.129)	(0.134)	(0.150)
signal$_{t-1}$		0.0318		0.0780
		(0.134)		(0.163)
signal$_{t-2}$				0.125
				(0.139)
Observations	677,120	677,120	545,240	545,240
R^2	0.849	0.849	0.857	0.857
Sample	$t-1$	$t-1$	$t-2$	$t-2$
Firm–country FE	Yes	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in t + 1, thus units are percentage points. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

with respect to the signal by the construction of the Export Compass.[5] Column (2) interacts the signal with past export status. The signal main effect becomes insignificant and falls to roughly one eighth of its column (1) value, while the interaction term is large and precisely estimated: firms already serving the destination respond by about 4.2 percentage points, whereas firms not yet serving it show no measurable response. Since incumbent firm–destination pairs are a small minority of the sample, the pooled estimate of 0.30 in column (1) is essentially this large effect diluted by a mass of non-responding observations.

Columns (3) and (4) restrict the sample to observations with no past export status to the destination, so that the estimates capture export market *entry* rather than the intensive margin. This restriction retains 93.6% of the full sample, confirming that only a small share of firm–destination–year observations already involve exporting to that destination. Column (3) mirrors column (2) in finding no measurable effect of a signal on entry. Column (4) replaces the signal indicator with a "thick" signal that equals one only if at least five buyers are listed, thereby discarding thin signals that name only one or two buyers. The entry effect now becomes significant at the 5% level, with a small but measurable magnitude of roughly one third of the main effect in Table 5. This pattern suggests that firms need to be presented with several potential buyers before entering a new market, consistent with higher fixed costs of entry into untapped destinations.

[5]Buyer information and contemporaneous export status are published in the same volume, so any change in export behaviour induced by the Export Compass should first appear in the following edition, unless a firm obtained the buyer information independently and at least a few months before the relevant volume was published.

Table 8: Past export status and export market entry

	(1)	(2)	(3)	(4)
			export_{t+1}	
signal_t	0.297***	0.0385	0.0306	
	(0.100)	(0.0882)	(0.0368)	
export_t	29.0***	28.6***		
	(1.23)	(1.23)		
$\text{signal}_t \times \text{export}_t$		4.19***		
		(0.746)		
signal_t (thick)				0.0944**
				(0.0442)
Observations	934,240	934,240	874,098	874,098
R^2	0.829	0.829	0.463	0.463
Sample	Full	Full	$\text{export}_t = 0$	$\text{export}_t = 0$
Firm–country FE	Yes	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in $t + 1$, so units are percentage points. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.5 Other robustness checks

5.5.1 Product-specific market demand and signal

A leading threat to identification is the possibility of product-specific demand booms in the markets from which signals are received. The rotation of consular coverage is set by geographic and administrative considerations and is therefore unlikely to track demand for any particular product; conditional on a market being covered, however, the probability that a signal appears in it may still be correlated with a positive demand shock for the goods that signal concerns.

To address this, I control directly for product-specific market conditions using a second component of the *Export-Compass*: the *Specielle Artikel* sections, in which consular posts reported on individual commodities. I extract 426 such entries from the 1890–1894 volumes and their post-1894 continuations, covering 79 consular posts and 237 distinct product designations, and code each for whether the consul is urging Austro-Hungarian producers to enter or expand in that market. Detection is grammatical rather than sentiment-based: consular recommendations are written in Konjunktiv II (*wäre, könnte, dürfte, sollte*), the mood of the counterfactual recommendation, and this marker separates advice from mere description in a way a dictionary of positive terms does not. A short example, from Ancona in 1890, reads: "The popular Viennese pendulum clocks and the so-called Karlstein clocks could easily find a market here too, if they were cheaper and better adapted to local taste." Others additionally characterize competitive conditions and the standing of Austro-Hungarian firms in the local market, as in this report from Tripoli in 1892: "Domestic

industry has only a small share in the import of silk goods; this could certainly be raised considerably if the manufacturers concerned tended this market rather better — which would be all the easier since Austria-Hungary produces extremely fine and reasonably priced oriental silks, which could find a large market here." Matching these product designations to the buyer-list categories yields a signal for 65 of the 3,141 country–year–category cells; the coding agrees with hand coding in 87% of a validation sample, and the resulting measurement error attenuates the estimates below toward zero.

Table A.4 splits the signal by the content of the contemporaneous consular report on the same product market. Column (1) adds an indicator for the consul describing Austro-Hungarian goods as already dominant or established in that commodity; column (2) an indicator for the report being encouraging, i.e. recommending entry or expansion. Both indicators are defined only where a report exists, so the coefficient on signal_t is the effect for product markets that were reported on but lacked the feature in question, and the second row is the additional effect of having it. Firms respond more strongly to signals from markets where Austro-Hungarian producers were already established (a further 3.85 percentage points, against a baseline of 0.28) and to signals accompanied by an encouraging report (a further 1.03 points). Neither drives the headline estimate: signals from product markets with no established Austro-Hungarian presence, and signals unaccompanied by an encouraging report, remain significant at 0.28 and 0.27 percentage points respectively, close to the saturated specification in Table 5. The differentials should be read as descriptive rather than precisely estimated: the position measure agrees with hand coding in only 47% of a validation sample, and both rest on the small subset of firm–destination pairs for which a matched product report exists.

5.5.2 Firms with export destination data for at least three years

Appendix Table A.1 re-estimates the baseline on a sample restricted to firms reporting destination-level information in at least three years. This restriction addresses potential selection from the contraction of destination coverage after 1891: if firms with incomplete destination reporting differ systematically from consistently reported firms, the baseline estimates could partly reflect reporting rather than behavior. The results are close to the baseline throughout: the signal coefficient is 3.03 percentage points without fixed effects (column 1), 1.34 with country–year fixed effects (column 2), 0.303 once firm–country fixed effects are added (column 3), and 0.248 in the saturated specification (column 4), compared to 2.89, 1.31, 0.301, and 0.289 in the full sample. The restricted-sample estimates are, if anything, marginally larger in the first three columns; only the saturated estimate falls, by roughly a seventh, and it remains significant at the 5% level. Some attenuation there is unsurprising given that the restriction removes roughly a sixth of the observations and, by construction, the short-lived firms whose entries and exits contribute much of the identifying variation. The stability of the point estimates indicates that selective reporting does not drive the baseline results.

5.5.3 Continuous treatment

Appendix Table A.2 exploits the intensive margin of the information treatment: the number of product-matched buyers listed for a destination, rather than a binary coverage indicator. Column (1) shows that one hundred additional listed buyers raise the probability of exporting to that destination in the following year by 0.66 percentage points—about 0.0066 percentage points per buyer, so that a firm facing ten additional listed buyers is roughly 0.07 percentage points more likely to enter—conditional on current export status and the full set of pairwise fixed effects. The effect is precisely estimated despite the demanding specification, which absorbs all destination-year demand shocks through country–year fixed effects: identification comes from comparing firms with more versus fewer product-matched buyers within the same destination-year cell. Column (2) shows that the relationship is not an artefact of the largest listings, with a doubling of the number of matched buyers raising the entry probability by roughly 0.10 percentage points, and column (3) confirms that what matters is the total volume of matched buyers rather than the size of the single largest product signal: conditioning on the maximum count leaves the coefficient on the total intact, while the maximum itself enters negatively and insignificantly. Column (4) reports the placebo. If the estimated coefficient reflected unobserved firm–destination demand trends—firms expanding toward destinations where buyer demand was growing—then buyer counts published in the following year's *Export Compass*, which firms could not yet observe, should also predict entry. They do not: the coefficient on the one-year lead is −0.145 with a standard error of 0.182, statistically indistinguishable from zero and estimated precisely enough to exclude an effect anywhere near the contemporaneous one. Export behavior responds to buyer information only once it is published, consistent with the information channel and inconsistent with firms anticipating future demand conditions. The smaller sample in column (4) reflects the loss of the final panel year required to construct the lead.

6 Mechanisms

6.1 Importance of signal strength

In this section I examine signal strength, i.e. how many potential buyers firms observe. Figure 2 decomposes the signal into three bins based on the number of product-matched buyers. The dummy $\text{signal}_t^{1\text{–}20}$ equals one when a firm receives a signal consisting of 1–20 buyers from a given country, and analogously for the 20–60 and 60+ bins; the omitted category is receiving no signal at all. The ex ante expectation is that stronger signals convey more information and should therefore make finding a buyer easier, since more listed importers mean more potential counterparties to approach. The main countervailing force is the novelty of information: destinations with many listed importers may be markets where demand for these products was already well known, or where consular offices had been active for years, so the marginal value of the listing may be lower.

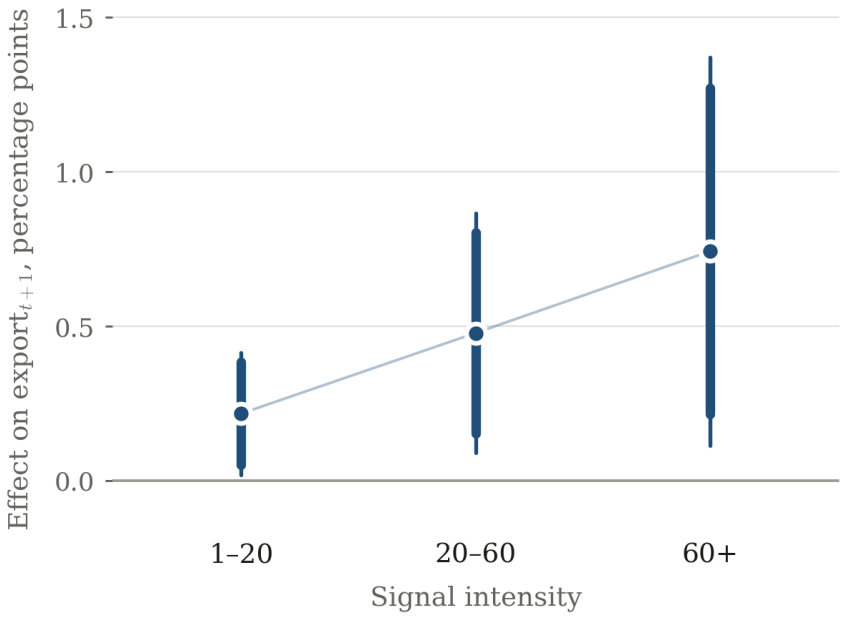


Figure 2: Effect of signal intensity

The results show a clear dose-response pattern. All three coefficients are positive, and they increase monotonically in signal strength: weak signals of 1–20 buyers raise the probability of exporting to the destination in the following year by 0.19 percentage points, medium signals of 20–60 buyers by 0.49 percentage points, and strong signals of more than 60 buyers by 0.62 percentage points—roughly three times the weak-signal effect. The weak-signal coefficient is close to the baseline estimate, which is expected given that most signals fall in this bin, while the medium bin is the most precisely estimated. Precision declines for the strongest signals, reflecting the smaller number of destination-years with more than 60 matched buyers, so the 60+ coefficient is significant only at the 10% level. Two conclusions follow. First, even weak signals matter: identifying a handful of potential buyers is enough to induce entry relative to no information at all, which speaks against a pure salience interpretation and is consistent with the low per-buyer cost of approaching a listed counterparty. Second, the monotonic gradient confirms, in a flexible form, the linear dose-response result from the continuous specification: the effect of information scales with its content, not merely its presence. The concern that abundant listings signal already-saturated markets finds no support in the data—if anything, the largest signals produce the largest responses, suggesting that the demand information dominated any loss of novelty.

6.2 Information Effects Across Destination Markets

Table 9 reports heterogeneity in the effect of buyer signals across destination-country characteristics, focusing on geography. Destinations are grouped into four categories: whether a country is *overseas* and therefore reached across an ocean, whether it shares a border with Austria-Hungary

(*neighbour*), whether it lies on the Balkan peninsula with established trade links (*balkan*), and whether it is located around the Mediterranean and readily accessible from the ports of Trieste and Fiume (*mediterranean*). These categories proxy for transport costs, distance, and the practical ease of serving a market.

The results suggest that the usefulness of buyer information depended on how actionable it was. In column (1), the baseline signal effect for non-overseas destinations is 0.36 percentage points and statistically significant, while the interaction with the overseas indicator is negative and of a similar order of magnitude (−0.271). Although the interaction is not statistically significant, the point estimates imply that the export response to signals from overseas markets was close to zero. This is consistent with firms being less able to act on information from distant destinations, likely because high fixed costs, transport frictions, and limited prior commercial presence made entry or reallocation more difficult, especially measure within a year. Furthermore, expected costs for importers might have prevented them entering business with distant and less-known exporters.

Table 9: Heterogeneity by destination country characteristics

	(1)	(2)	(3)	(4)
	export_{t+1}	export_{t+1}	export_{t+1}	export_{t+1}
signal_t	0.360**	0.195**	0.270***	0.112
	(0.159)	(0.0844)	(0.0921)	(0.135)
$\text{signal}_t \times \text{overseas}_t$	−0.271			
	(0.186)			
$\text{signal}_t \times \text{neighbour}_t$		0.292		
		(0.302)		
$\text{signal}_t \times \text{balkan}_t$			−0.0196	
			(0.353)	
$\text{signal}_t \times \text{mediterranean}_t$				0.446**
				(0.210)
Observations	934,240	934,240	934,240	934,240
R^2	0.828	0.828	0.828	0.828
Firm–country FE	Yes	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in t + 1, thus units are percentage points. The interaction categories are defined as follows: *overseas* denotes countries on other continents reached across an ocean; *neighbour* denotes countries bordering Austria-Hungary; *balkan* denotes countries with established trade links on the Balkan peninsula; and *mediterranean* denotes countries around the Mediterranean Sea readily accessible from Trieste or Fiume. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The interaction for neighboring countries in column (2) points in the opposite direction. The coefficient is positive and economically sizeable—the implied total effect, at roughly 0.292, is more

than double the effect elsewhere—but it is imprecisely estimated and cannot be distinguished from zero. The direction is nevertheless plausible: neighboring destinations were easier to serve because of shorter transport distances, lower transaction costs, and denser pre-existing commercial links, thus buyer information in such markets was more likely to translate into actual export entry or reallocation.

For Balkan destinations, column (3) shows an interaction that is rather negative. Signals from these markets were less effective than signals from elsewhere. This is rather surprising, since Balkan countries were important destinations for Austro-Hungarian exports. Even thought information from these countries is actionable, likely its novelty is less significant since the Austro-Hungarian Empire invested in uncovering business potential for the Balkan countries even before the lists for potential buyers were published in the Export Compass.

The clearest differential effect appears for Mediterranean destinations in column (4). The interaction is positive and statistically significant at the 5% level (0.446), while the baseline effect for non-Mediterranean markets is smaller and insignificant. The implied total effect for Mediterranean destinations, roughly 0.558, is about twice the baseline estimate reported in Table 5. This pattern is plausible given that these markets were connected to the monarchy through established maritime routes from Trieste and Fiume, which kept the cost of acting on buyer information low; at the same time, they may have been less saturated with existing commercial ties than immediate neighbors, leaving more scope for new matches.

Overall, Table 9 shows that the effect of buyer signals was concentrated in destinations where information could be turned into action at relatively low cost—most clearly the Mediterranean markets served through the monarchy's own ports, and, in terms of point estimates, neighboring countries—while it was muted for distant overseas destinations. The pattern therefore supports an interpretation of export promotion as reducing search and matching frictions conditional on market accessibility, rather than as a uniform information shock across all destinations.

6.3 The role of buyer types: intermediaries (merchants) and nationality

Table 10 shows the results from the heterogeneity analysis concerning buyer characteristics. The merchant share of a signal is calculated as $N_{\text{merchant}}/N_{\text{all}}$, i.e. the ratio of merchants to all buyers. As indicated above, those buyers are considered merchants that either purchase en gross (in large quantities) or directly indicate trader status (e.g. "Händler" or "Kommissionär"). Similarly, the Austro-Hungarian share is calculated as $N_{\text{Austro-Hungarian}}/N_{\text{all}}$, i.e. the ratio of Austro-Hungarian buyers to all buyers. The dummies *Merchant share high* and *Austro-Hungarian share high* are defined in two ways: for shares above 0.5 (columns 1 and 3) and for shares above the sample median (columns 2 and 4). Whether a buyer is Austro-Hungarian is explicitly indicated in the list of buyers, and their categorization therefore required no further assumptions. Ex ante, a higher

share of intermediary buyers should increase the probability of exporting to a given destination, since intermediaries purchase goods in larger quantities and are likely to have better information about further downstream demand. The role of Austro-Hungarian buyers is more ambiguous: on the one hand, exporters were more likely to already know these firms, so listing them may convey less novel information. On the other hand, establishing new commercial connections is easier with firms that operate under a shared legal framework, and communication should be smoother due to a common language, culture, and customs.

Table 10: The role of intermediaries and common citizenship for buyers

	(1)	(2)	(3)	(4)
	export_{t+1}	export_{t+1}	export_{t+1}	export_{t+1}
signal_t	0.255**	0.213*	0.291***	0.318***
	(0.106)	(0.111)	(0.106)	(0.0997)
$\text{signal}_t \times$ high merchant share_t	0.643***			
	(0.231)			
$\text{signal}_t \times$ high merchant share $(\text{med.})_t$		0.253*		
		(0.143)		
$\text{signal}_t \times$ high Austro-Hung. share_t			−0.153	
			(0.781)	
$\text{signal}_t \times$ high Austro-Hung. share $(\text{med.})_t$				−0.119
				(0.176)
Observations	934,240	934,240	934,240	934,240
R^2	0.818	0.818	0.818	0.818
Firm–country FE	Yes	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in $t+1$, thus units are percentage points. Columns (1) and (3) define high-share as more than 50% of buyers within a signal, while columns (2) and (4) use the sample median as the threshold. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The results confirm the ex ante expectation about merchant share. In column (1), when the merchant-share dummy equals zero, the signal effect is 0.255 percentage point increase, in line with the baseline findings. When the merchant-share dummy equals one, the total effect rises by approximately 0.643—a roughly threefold increase—indicating that signals dominated by wholesale intermediaries are substantially more informative for export entry decisions. The pattern is robust to the choice of threshold: with the median cut in column (2), the interaction remains positive and significant at the 10% level, and the implied total effect (about 0.466 percentage points increase) is more than double the effect of low-merchant-share signals, which remains significant on the 10% level. These results support the interpretation that firms buying in large quantities, as well as firms specializing in international procurement, meaningfully facilitating export market presence. They likely maintain closer ties with local consular offices and may actively seek out potential foreign

suppliers.

The evidence on Austro-Hungarian buyers is far less conclusive. In column (3), the interaction coefficient is negative and economically meaningful, however, it is imprecisely estimated and statistically insignificant, partially due to the very low number of signals containing high Austro-Hungarian share. Under the median threshold in column (4), the coefficient of the interaction terms is similar to the results in column (3). The two definitions thus give no consistent indication that signals dominated by Austro-Hungarian buyers were differentially effective. A natural reading is that the two ex ante forces offset each other: any advantage from contract enforceability and cultural–linguistic familiarity may have been neutralized by the lower novelty of information about compatriot firms that exporters were more likely to know already. Given the imprecision of the estimates, however, this interpretation should be treated with caution.

Table A.3 replaces the discretized dummies with the underlying continuous shares, which enter the specification directly rather than interactively. The coefficients are therefore not marginal effects at a threshold but the change in the export-entry probability associated with moving from a signal in which no buyer is of the relevant type to one in which every buyer is. Columns (1) and (2) introduce the two shares one at a time, column (3) includes both jointly, and column (4) substitutes the level of demand with its logarithm; all specifications retain the full firm $\times$ country, firm $\times$ year, and country $\times$ year fixed effects.

The continuous measure corroborates the dummy results for intermediaries. In column (3), shifting the buyer composition of a signal from entirely non-merchant to entirely merchant raises the probability of exporting to the destination in $t+1$ by 0.700 percentage points, significant at the 1% level, and the estimate is essentially unchanged when the Austro-Hungarian share is omitted (0.693 in column (1)) or when demand enters in logs (0.585 in column (4)). Relative to the baseline signal effect of roughly 0.255 percentage points, this is an increase of comparable magnitude to the threefold effect implied by the dummy interaction in Table 10, which is reassuring given that the two specifications exploit different parts of the variation: the dummies contrast coarse groups, whereas the continuous measure uses the full distribution of the share and is thus the more demanding test. The demand terms themselves remain positive and precisely estimated—0.00607 percentage points per additional unit of demand in column (1) and 0.127 percentage points per log point in column (4)—so the merchant-share coefficient is not simply absorbing the size of the signal or the level of destination demand, which would otherwise be the most obvious confound, since larger signals mechanically contain more wholesale buyers.

The Austro-Hungarian share again yields no usable evidence. The point estimate is negative throughout, ranging from -0.183 percentage points in column (2) to -0.525 in column (4), but the standard errors are of the same order as the coefficients themselves and none of the estimates approaches conventional significance. The pattern is thus consistent with the dummy specifications: the sign is stable but the magnitude is not identified, and the confidence intervals comfortably contain both economically substantial negative effects and zero. Two features of the data caution

against reading the negative sign as informative. First, signals with a high Austro-Hungarian share are rare, so the upper range of the distribution is thinly populated. Second, shares constructed from a small number of buyers are measured with considerable noise, and the resulting attenuation biases the coefficient towards zero—a concern that applies to both shares but bites hardest where the underlying counts are smallest. The joint reading of the two tables is therefore that the intermediary channel is robust to how buyer composition is defined, whereas the nationality channel remains an open question that the present data cannot settle.

6.4 Stronger persistence with merchant connections

Table 11 shows the results for the difference between signals with low vs. high merchant shares, as defined in Section 6.3. Column (1) shows the baseline interaction that was already presented in Table 10. The differential is estimated precisely enough to be significant at the 1% level, and it is the interaction rather than the un-interacted term that carries most of the average effect documented in Table 5.

Table 11: Persistence of effect for high merchant signals

	(1)	(2)	(3)
		export_{t+1}	
signal_t	0.253**	0.254**	0.235*
	(0.105)	(0.110)	(0.138)
$\text{signal}_t \times \text{high merchant share}_t$	0.604***	0.364*	0.379
	(0.224)	(0.213)	(0.262)
signal_{t-1}		−0.0827	−0.163
		(0.133)	(0.135)
$\text{signal}_{t-1} \times \text{high merchant share}_{t-1}$		0.483**	0.409*
		(0.242)	(0.222)
signal_{t-2}			0.109
			(0.128)
$\text{signal}_{t-2} \times \text{high merchant share}_{t-2}$			0.734***
			(0.278)
Observations	934,240	604,200	415,080
R^2	0.819	0.864	0.899
Firm–country FE	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes

Notes: Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in $t + 1$, thus units are percentage points. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Columns (2) and (3) add lagged signals, interacted with the merchant-share indicator measured at the same date. The un-interacted lags are small and statistically indistinguishable from zero (0.02 and −0.03 percentage points at $t-1$; 0.18 at $t-2$), consistent with the results in Table 7. The interaction terms, by contrast, remain positive at every horizon—0.39 percentage points at $t-1$ and 0.81 at $t-2$ in column (3)—so that the total effect for high-merchant-share destinations is 0.59, 0.36 and 0.98 percentage points for signals received at t, $t-1$ and $t-2$ respectively. Two features of the table caution against a stronger reading. First, the samples differ across columns (934,240, 604,200 and 415,080 observations) because each additional lag requires an additional year of observed signal history, so the attenuation of the contemporaneous interaction from 0.64 to 0.34 cannot be separated from the change in sample composition. Second, the confidence intervals on the interactions are wide relative to the differences between them, reflecting the low prevalence of the moderator. Overall, signals with high share of merchants specializing in wholesale seem to result in stronger commercial connections that persist well beyond one year. The difference between signals with low vs. high merchant shares are clear even if the sample is reduced to half of its original size.

7 Firm-level outcomes

Table 12 aggregates the analysis to the firm-year level to ask whether buyer information changed firms' overall export portfolios or merely redirected them. Two treatment measures distinguish the breadth of information from its depth: the number of destinations from which a firm received a signal (*signaled destinations*), and the total number of matched buyers summed across all destinations (*total matched buyers*). Outcomes are the number of export destinations and the number of exported products in the following year; all specifications control for the initial outcome and include firm and year fixed effects, so identification comes from within-firm changes in information exposure over time. Three findings emerge. First, the geographic scope of exporting responds to the depth of information, not its breadth: the count of signaled destinations has no significant effect on next-year destination count (column 1), but the total number of matched buyers does (column 2)—one hundred additional listed buyers raise the destination count by 0.056, relative to a sample mean of roughly 2.5 destinations. This mirrors the dyadic results: what drives entry is the volume of actionable buyer information, not the mere fact of coverage. Second, because total destination counts rise with information, the destination-level entry effects do not reflect pure reallocation across markets: buyer information expanded firms' export portfolios on net rather than merely redirecting sales from non-signaled to signaled destinations. Third, information breadth is associated with product specialization: firms receiving signals from more destinations reduce their exported product range (column 3), consistent with firms concentrating on product lines for which foreign demand has been identified, while product scope does not respond to buyer depth (column 4). These firm-level results should be read as a complement to, not a substitute for, the dyadic analysis: aggregation to the firm-year level forgoes the within-destination-year comparisons that

identify the main results, and the estimates rely on parallel trends across firms with different information exposure. With that caveat, the pattern is coherent: public buyer information led firms to serve more markets with a narrower, demand-matched product range.

Table 12: Firm-level outcomes

	(1)	(2)	(3)	(4)
	dest_count$_{t+1}$		product_num$_{t+1}$	
signal count	0.00985		−0.0873***	
	(0.00967)		(0.0318)	
firms sum/100		0.0516***		0.0726
		(0.0164)		(0.0468)
Observations	26,978	26,978	18,088	18,088
R^2	0.621	0.621	0.714	0.714
Initial outcome	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

8 Conclusion

The findings outlined in the previous sections carry a message for the evaluation of export promotion policies. The Export Compass was not a subsidy or a logistical intervention; it was a relatively cheap information service. Since it generated measurable results for export market participation, it suggests that not knowing *whom* to sell to was a first-order barrier to trade—and the pattern of effects suggests why modern experimental evaluations often find no extensive-margin response: what moved firms was not generic market information but specific, actionable buyer identities, delivered repeatedly and at scale. Several extensions are left for future work, including within the scope of this project: linking buyer information to firm innovation and product upgrading, and to firm longevity—whether market intelligence fostered durable participation in international trade or merely temporary entry. More broadly, the setting offers what modern administrative data cannot: direct observation of the information itself. Opening this black box further remains a promising avenue for understanding how buyer–seller matches form.

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A Other tables

Table A.1: Effect of buyer-information signals on next-period exporting for firms with destination info for at least 3 years

	(1)	(2)	(3)	(4)
	export$_{t+1}$			
signal$_t$	3.00***	1.27***	0.254**	0.237**
	(0.153)	(0.228)	(0.119)	(0.108)
Observations	776,200	776,200	776,200	776,200
R^2	0.001	0.129	0.820	0.841
Firm–country FE	No	No	Yes	Yes
Firm–year FE	No	No	No	Yes
Country–year FE	No	Yes	Yes	Yes

Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in t + 1, thus units are percentage points. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.2: Buyer counts as continuous measure

	Dependent variable: Export$_{t+1}$			
	(1)	(2)	(3)	(4)
Buyers$_t$ (/100)	0.613***		0.987**	
	(0.228)		(0.484)	
log Buyers$_t$		0.132***		
		(0.0459)		
Max buyers$_t$ (/100)			−0.621	
			(0.699)	
Placebo: Buyers$_{t+1}$ (/100)				−0.337*
				(0.197)
Observations	934,240	934,240	934,240	723,520
R^2	0.819	0.819	0.819	0.912
Firm–country FE	Yes	Yes	Yes	Yes
Firm–year FE	Yes	Yes	Yes	Yes
Country–year FE	Yes	Yes	Yes	Yes

Notes: The dependent variable is an indicator for exporting to destination d in year $t+1$. *Buyers* is the number of product-matched buyers listed in the Compass for destination d in year t, divided by 100; thus, the coefficient measures the change in the probability of exporting associated with an additional 100 listed buyers. Column (2) reports the log version of column (1). Column (3) adds the number of max buyers for the largest product signal. Column (4) replaces column (1) with its one-year lead, i.e. buyers listed in the *following* year's edition, which was not yet available to firms at time t, as a placebo test. Standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.3: Buyer counts as continuous measure

	(1)	(2)	(3)	(4)
	Dependent variable: export$_{t+1}$			
Demand sum (×100)	0.00607***	0.00613***	0.00608***	
	(0.00228)	(0.00228)	(0.00227)	
Merchant share	0.00693***		0.00700***	0.00585**
	(0.00246)		(0.00243)	(0.00246)
Austro-Hungarian share		−0.00183	−0.00253	−0.00525
		(0.00635)	(0.00632)	(0.00633)
Log demand sum				0.00127***
				(0.000464)
Observations	934,240	934,240	934,240	934,240
R-squared	0.819	0.819	0.819	0.819
Firm × country FE	Yes	Yes	Yes	Yes
Firm × year FE	Yes	Yes	Yes	Yes
Country × year FE	Yes	Yes	Yes	Yes

Notes: The dependent variable is an indicator for exporting to destination d in year $t+1$. *Buyers* is the number of product-matched buyers listed in the Compass for destination d in year t, divided by 100; thus, the coefficient measures the change in the probability of exporting associated with an additional 100 listed buyers. Column (2) reports the log version of column (1). Column (3) adds the number of max buyers for the largest product signal. Column (4) replaces column (1) with its one-year lead, i.e. buyers listed in the *following* year's edition, which was not yet available to firms at time t, as a placebo test. Standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.4: Market reports and their interaction with buyer signals

	(1)	(2)
	export_{t+1}	export_{t+1}
signal_t	0.277***	0.274***
	(0.104)	(0.104)
$\text{signal}_t \times \text{AH dominant}_t$	3.85**	
	(1.56)	
$\text{signal}_t \times \text{encouraging}_t$		1.03**
		(0.448)
Observations	934,240	934,240
R^2	0.819	0.819
Firm–country FE	Yes	Yes
Firm–year FE	Yes	Yes
Country–year FE	Yes	Yes

Notes: *Notes:* Coefficients and standard errors multiplied by 100; the dependent variable is an indicator for exporting to a destination in t + 1, thus units are percentage points. Column (1) adds an indicator for the report describing Austro-Hungarian goods as dominant or established and column (2) for an encouraging tone. These indicators are nested within $signal_t$, so their coefficients are differentials relative to other reported categories and should not be added to it. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

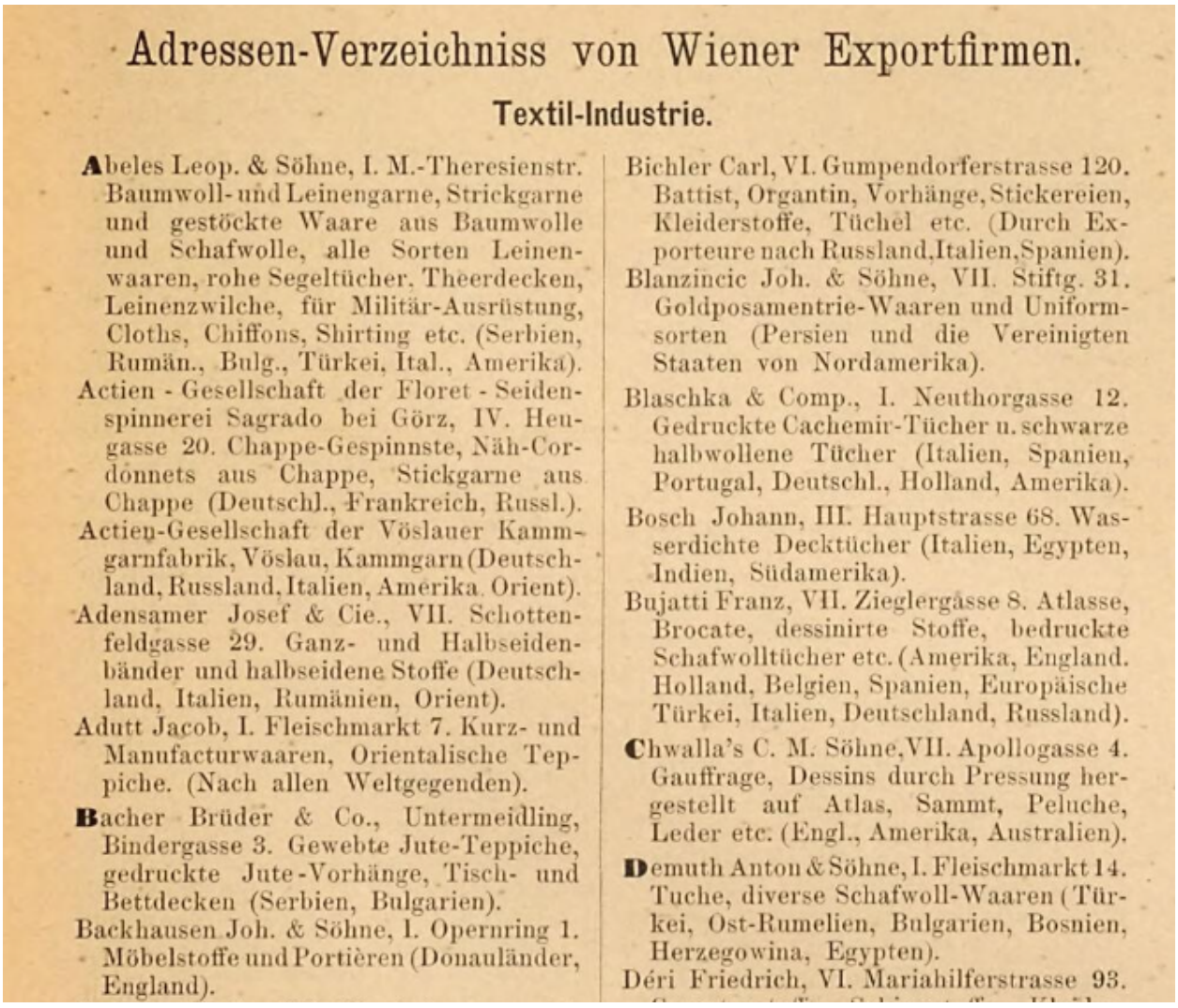
B Other figures



Figure B.1: Map of consular offices in Europe



Figure B.2: Map of consular offices in South America



Adressen-Verzeichniss von Wiener Exportfirmen.

Textil-Industrie.

Abeles Leop. & Söhne, I. M.-Theresienstr. Baumwoll- und Leinengarne, Strickgarne und gestöckte Waare aus Baumwolle und Schafwolle, alle Sorten Leinenwaaren, rohe Segeltücher, Theerdecken, Leinenzwilche, für Militär-Ausrüstung, Cloths, Chiffons, Shirting etc. (Serbien, Rumän., Bulg., Türkei, Ital., Amerika).

Actien - Gesellschaft der Floret - Seidenspinnerei Sagrado bei Görz, IV. Heugasse 20. Chappe-Gespinnste, Näh-Cordonnets aus Chappe, Stickgarne aus Chappe (Deutschl., Frankreich, Russl.).

Actien-Gesellschaft der Vöslauer Kammgarnfabrik, Vöslau, Kammgarn (Deutschland, Russland, Italien, Amerika, Orient).

Adensamer Josef & Cie., VII. Schottenfeldgasse 29. Ganz- und Halbseidenbänder und halbseidene Stoffe (Deutschland, Italien, Rumänien, Orient).

Adutt Jacob, I. Fleischmarkt 7. Kurz- und Manufacturwaaren, Orientalische Teppiche. (Nach allen Weltgegenden).

Bacher Brüder & Co., Untermeidling, Bindergasse 3. Gewebte Jute-Teppiche, gedruckte Jute-Vorhänge, Tisch- und Bettdecken (Serbien, Bulgarien).

Backhausen Joh. & Söhne, I. Opernring 1. Möbelstoffe und Portièren (Donauländer, England).

Bichler Carl, VI. Gumpendorferstrasse 120. Battist, Organtin, Vorhänge, Stickereien, Kleiderstoffe, Tüchel etc. (Durch Exporteure nach Russland, Italien, Spanien).

Blanzincic Joh. & Söhne, VII. Stiftg. 31. Goldposamentrie-Waaren und Uniformsorten (Persien und die Vereinigten Staaten von Nordamerika).

Blaschka & Comp., I. Neuthorgasse 12. Gedruckte Cachemir-Tücher u. schwarze halbwollene Tücher (Italien, Spanien, Portugal, Deutschl., Holland, Amerika).

Bosch Johann, III. Hauptstrasse 68. Wasserdichte Decktücher (Italien, Egypten, Indien, Südamerika).

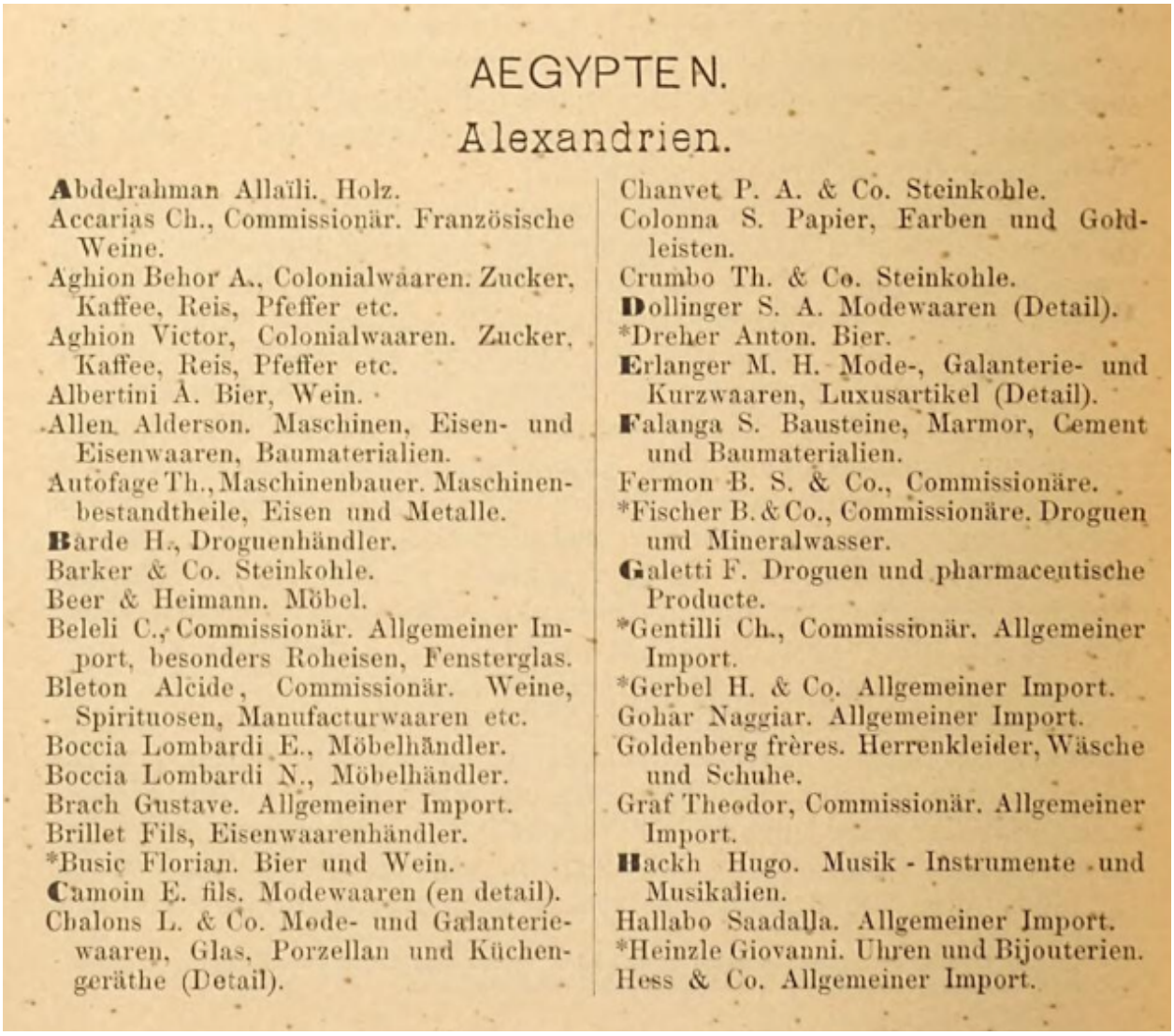
Bujatti Franz, VII. Zieglergasse 8. Atlasse, Brocate, dessinirte Stoffe, bedruckte Schafwolltücher etc. (Amerika, England. Holland, Belgien, Spanien, Europäische Türkei, Italien, Deutschland, Russland).

Chwalla's C. M. Söhne, VII. Apollogasse 4. Gauffrage, Dessins durch Pressung hergestellt auf Atlas, Sammt, Peluche, Leder etc. (Engl., Amerika, Australien).

Demuth Anton & Söhne, I. Fleischmarkt 14. Tuche, diverse Schafwoll-Waaren (Türkei, Ost-Rumelien, Bulgarien, Bosnien, Herzegowina, Egypten).

Déri Friedrich, VI. Mariahilferstrasse 93.

Figure B.3: Example of raw data from exporting firms



AEGYPTEN.

Alexandrien.

Abdelrahman Allaïli. Holz.
Accarias Ch., Commissionär. Französische Weine.
Aghion Behor A., Colonialwaaren. Zucker, Kaffee, Reis, Pfeffer etc.
Aghion Victor, Colonialwaaren. Zucker, Kaffee, Reis, Pfeffer etc.
Albertini A. Bier, Wein.
Allen Alderson. Maschinen, Eisen- und Eisenwaaren, Baumaterialien.
Autofage Th., Maschinenbauer. Maschinenbestandtheile, Eisen und Metalle.
Barde H., Droguenhändler.
Barker & Co. Steinkohle.
Beer & Heimann. Möbel.
Beleli C., Commissionär. Allgemeiner Import, besonders Roheisen, Fensterglas.
Bleton Alcide, Commissionär. Weine, Spirituosen, Manufacturwaaren etc.
Boccia Lombardi E., Möbelhändler.
Boccia Lombardi N., Möbelhändler.
Brach Gustave. Allgemeiner Import.
Brillet Fils, Eisenwaarenhändler.
*Busic Florian. Bier und Wein.
Camoin E. fils. Modewaaren (en detail).
Chalons L. & Co. Mode- und Galanteriewaaren, Glas, Porzellan und Küchengeräthe (Detail).
Chanvet P. A. & Co. Steinkohle.
Colonna S. Papier, Farben und Goldleisten.
Crumbo Th. & Co. Steinkohle.
Dollinger S. A. Modewaaren (Detail).
*Dreher Anton. Bier.
Erlanger M. H. Mode-, Galanterie- und Kurzwaaren, Luxusartikel (Detail).
Falanga S. Bausteine, Marmor, Cement und Baumaterialien.
Fermon B. S. & Co., Commissionäre.
*Fischer B. & Co., Commissionäre. Droguen und Mineralwasser.
Galetti F. Droguen und pharmaceutische Producte.
*Gentilli Ch., Commissionär. Allgemeiner Import.
*Gerbel H. & Co. Allgemeiner Import.
Gohar Naggiar. Allgemeiner Import.
Goldenberg frères. Herrenkleider, Wäsche und Schuhe.
Graf Theodor, Commissionär. Allgemeiner Import.
Hackh Hugo. Musik - Instrumente und Musikalien.
Hallabo Saadalla. Allgemeiner Import.
*Heinzle Giovanni. Uhren und Bijouterien.
Hess & Co. Allgemeiner Import.

Figure B.4: Example of raw data from importing firms



Figure B.5: Matching importers and exporters through relevant products

C Data Construction

C.1 Source Data

The underlying data source is the *Austro-Hungarian Export Compass* (*Österreichisch-Ungarischer Export-Compass*), a commercial directory published annually listing exporting firms across the Habsburg Monarchy. I use the editions from 1890 to 1898. The raw data was digitized into Excel spreadsheets, with each row containing a single firm entry as a free-text string. These entries typically combine the firm's name, location or address, product descriptions, and miscellaneous information (e.g., legal form, advertisements, agency roles) in a single unstructured line of German-language text.

The data construction pipeline consists of six steps, summarized in Figure C.1 and described in detail below.

Figure C.1: Data Construction Pipeline

Step	Description
1	Parsing raw firm entries into structured fields
2	Product category classification
3	Export destination standardization
4	Cross-year firm matching
5	Panel assembly and firm characteristic merge
6	Firm–country–year long panel and import demand matching

C.2 Step 1: Parsing Firm Entries into Structured Fields

Given the heterogeneous and historically idiosyncratic formatting of the entries, manual rule-based parsing proved infeasible at scale. I employ a large language model (LLM)—specifically OpenAI's GPT-5.1—via the OpenAI API to decompose each unstructured firm entry into four structured fields:

1. **Name**: the firm name, including legal form (e.g., "& Comp.", "A.G.", "Gebrüder").
2. **Address**: the location, street, district, or regional designator (e.g., "Bez. Krumau", "Mähren").
3. **Product**: the goods or product categories associated with the firm.
4. **Other information**: residual content such as references to advertisements, speciality notes, or agency relationships.

An additional binary flag (`parse_ok`) indicates whether the entry was successfully recognized as a genuine firm listing, as opposed to an advertisement or editorial comment.

Each raw text entry is first cleaned by stripping whitespace and collapsing multiple spaces. Entries are then grouped into batches of 40 and submitted to the LLM in a single API call. The model receives a detailed German-language system prompt instructing it to separate each entry into the four target fields, with explicit rules about field assignment—for instance, that geographic qualifiers belong in the address field rather than the product field, and that product information can

be inferred from the firm type when not explicitly stated (e.g., a "Zuckerfabrik" produces sugar). The API temperature parameter is set to 0, ensuring deterministic output for reproducibility.

Each unique input text is hashed (SHA-1) and the corresponding parsed result is stored in a persistent JSON cache. This avoids redundant API calls when re-running the pipeline and provides robustness against interruptions. Failed API calls are retried up to three times with exponential back-off.

C.3 Step 2: Product Category Classification

The product descriptions extracted in Step 1 are free-text strings that frequently list multiple goods separated by commas, semicolons, or the conjunction "und" (e.g., "Colonialwaaren, Eisen und Glas"). A second pipeline assigns each firm one or more standardized product categories drawn from a controlled German-language vocabulary.

Each product string is first split into individual tokens using commas, semicolons, slashes, ampersands, and the word "und" as delimiters. All unique tokens across the dataset are collected, and those not yet cached are submitted in batches of 20 to GPT-5.1. The model receives the current list of existing categories and is instructed to assign each token to exactly one category, creating a new category if no suitable one exists. The prompt explicitly prohibits catch-all labels such as "Sonstiges" (miscellaneous) and encourages fine-grained distinctions—for instance, distinguishing "Baumwollwaren" (cotton goods) from "Seidenwaren" (silk goods) rather than grouping them under a generic "Textilien" label. Similarly, colonial goods are disaggregated into specific commodities (coffee, sugar, tea, tobacco, etc.).

A deterministic post-processing step merges known near-synonyms arising from historical spelling variation. For example, "Colonialwaaren" and "Kolonialwaren" are unified, as are "Eisenwaaren" and "Eisenwaren," reflecting the orthographic reform from *aa* to *a* that occurred in this period. The category vocabulary is initialized with a seed set of approximately 30 categories spanning major sectors (beverages, foodstuffs, textiles, metals, machinery, etc.) and grows organically as the model encounters new product types. As in Step 1, all token-to-category mappings are cached for reproducibility.

C.4 Step 3: Export Destination Standardization

The Export Compass records not only the goods each firm traded but also its export destinations, recorded as free-text strings listing countries, regions, and abbreviated place names in the conventions of the 1890s (e.g., "Ital.", "Rumän.", "Orient"). A third pipeline standardizes these entries into a controlled vocabulary of country and region names.

Each destination string is split on commas, semicolons, slashes, dashes, and the conjunction "und" into individual tokens. Stopwords and noise tokens (e.g., "etc.", "via", "über") are discarded. Before invoking the LLM, a fast deterministic lookup resolves common abbreviations and spelling variants (e.g., "Ital." → "Italien", "Holland" → "Niederlande", "England" → "Großbritannien"). Tokens that match an existing category after Unicode normalization and case-folding are also resolved locally, avoiding unnecessary API calls.

Remaining unresolved tokens are submitted in batches of 10 to GPT-5.1. The model receives the current list of standardized categories and is instructed to map each input to exactly one existing category or, if none fits, to create a new one. The prompt explicitly forbids composite categories (e.g., "Deutschland und Schweiz") and catch-all labels, and instructs the model to stay as close as possible to the literal text—mapping "Bulgarien" to "Bulgarien" rather than abstracting it to "Balkan."

The category list is initialized with a curated set of approximately 60 entries covering the major countries and macro-regions relevant to late-nineteenth-century Austro-Hungarian trade (e.g., European states, Ottoman territories, colonial destinations, and broad regional labels such as "Orient," "Levante," and "Überseeisch"). After standardization, each firm's destination list is split into two sub-lists: specific countries (e.g., "Serbien," "Rumänien") and broad regions (e.g., "Balkan," "Südamerika," "Weltweit"), based on a predefined set of region labels.

C.5 Step 4: Cross-Year Firm Matching

To track firms over time and construct a panel dataset, firm entries are linked across all available years (1890–1898) using a combination of exact and fuzzy name matching.

Firm names are preprocessed by stripping diacritics, lowercasing, removing punctuation, expanding abbreviations (e.g., "&" → "und"), and stripping legal suffixes and corporate designators (e.g., "A.G.", "Ges.m.b.H.", "Gebrüder", "& Co.", "Fabrik"). To reduce the computational burden, firms are grouped into blocks defined by the first two characters of the normalized name. Candidate matches are drawn primarily from the same block; if no suitable candidate is found, the full pool is searched as a fallback.

For each pair of years, firms whose normalized names are identical are matched first (exact match, score = 100). Remaining unmatched firms are compared using the token-set-ratio metric from the RapidFuzz library, which is robust to word reordering and partial overlap. A match is accepted only if the similarity score meets or exceeds a threshold of 90. A match between firm i in year t and firm j in year t' is accepted only if j is i's highest-scoring candidate *and* i is j's highest-scoring candidate—a mutual (symmetric) best-match criterion that guards against false positives.

All accepted pairwise links across all $\binom{9}{2} = 36$ year-pairs are fed into a Union-Find (disjoint-set) data structure. This groups firm observations that are transitively linked—if firm i in 1890 matches firm j in 1892, and j matches firm k in 1894, all three are assigned to the same cluster. Each cluster receives a stable panel identifier (`ex_uid`).

C.6 Step 5: Panel Assembly and Firm Characteristic Merge

The pairwise match edges from Step 4 are represented as an undirected graph using the NetworkX library, where each node is a year–firm pair and each edge is an accepted match link. Connected components identify firm clusters, each assigned a stable panel identifier.

The graph is converted into a long-format panel with one row per firm-year observation. A wide-format version is also produced, with one row per firm cluster and separate columns for the firm's original ID in each year. The long panel is then merged with the enriched yearly firm files produced by Steps 1–3, attaching the full set of firm-level characteristics—parsed name, address, product descriptions, standardized product categories, export destinations, and standardized country and region classifications—to each panel observation.

C.7 Step 6: Firm–Country–Year Panel and Import Demand Matching

The final analytical dataset is constructed by reshaping the panel into a firm–country–year structure and merging it with external data on import demand.

A master list of all destination countries appearing anywhere in the data is compiled. Each firm-year observation is crossed with the full set of countries, producing a row for every possible firm–country–year combination. A binary indicator is set to 1 if the firm lists that country among its export destinations in that year, and 0 otherwise. This structure enables gravity-style estimation of both the intensive and extensive margins of trade.

To measure the demand environment each firm faces in each destination, the dataset is further exploded along the product dimension: each firm–country–year row is expanded into firm–country–year–category rows using the standardized product categories from Step 2. These are then merged with an external file recording the number of importing firms by country, year, and product category—a proxy for destination-specific import demand. The product-level demand matches are aggregated back to the firm–country–year level, producing summary measures: the sum, mean, and maximum of matched import demand across the firm's product categories, as well as the count of categories for which a demand match was found.

The resulting dataset—one row per firm–country–year with export indicators, firm characteristics, and demand measures—is exported both as an Excel file and as a Stata `.dta` file.

C.8 Validation

Entries flagged with `parse_ok = false` by the model in Step 1 were reviewed manually. A random sample of successfully parsed entries was also inspected to verify the quality of field separation. For the product categorization in Step 2, category frequency distributions were examined to detect overly broad or inconsistent labels. In Step 4, match quality was assessed through the distribution of fuzzy match scores and the share of exact versus fuzzy matches across year-pairs. The deterministic (zero-temperature) setting and the caching mechanism in Steps 1–3 ensure that results are reproducible across runs.

D Descriptive statistics from entity resolution

D.1 Data Construction and Firm Matching

The dataset is constructed by digitising the Austro-Hungarian Export Compass for the years 1890–1898. Table D.1 reports the number of firm entries per edition after removing within-year duplicates. The raw entry count grows steadily from 3,255 in 1890 to 11,355 in 1898, reflecting both genuine market expansion and the increasing coverage of later editions. After deduplication the usable sample rises from 3,173 to 10,320 firm-year observations; duplicate rates range from 2.5% in 1890 to 9.1% in 1898.

Table D.1: Sample size by year

Year	Raw entries	After deduplication	Duplicates removed
1890	3,255	3,173	82
1891	5,533	5,331	202
1892	6,172	5,907	265
1893	7,679	7,061	618
1894	8,989	8,114	875
1895	9,605	8,727	878
1896	10,266	9,418	848
1897	11,025	10,074	951
1898	11,355	10,320	1,035

D.2 Cross-Year Firm Matching

To track firms across editions, each firm name in year t is matched against all firm names in every other year t' using a combination of exact string matching and fuzzy matching based on edit-distance scores normalised to $[0, 100]$. Across all 36 year pairs, a total of 151,708 cross-year matches are identified, of which 132,854 (87.6%) are exact and 18,854 (12.4%) are fuzzy. The mean match score is 99.36 and the median is 100, indicating that the vast majority of fuzzy matches involve only minor name variations (e.g. spelling differences or abbreviations) rather than substantive discrepancies.

Table D.2 reports match counts and quality for consecutive year pairs. The number of matched firms is naturally highest for adjacent years and declines as the gap widens, consistent with firm entry and exit over time. Match quality remains high throughout: mean scores for consecutive pairs all exceed 99.6.

D.3 Firm Persistence Across Editions

Table **??** reports the distribution of firms by the number of editions in which they are observed. The majority of firms—29.7%—appear in only a single edition, suggesting substantial turnover in the exporter population. At the other extreme, 1,513 firms (8.1%) are observed in all nine editions, forming a stable core of persistent exporters. Firms observed in two or three editions together account for a further 28.9% of the sample.

Table D.2: Cross-year matches for consecutive year pairs

Year pair	Total matches	Exact	Fuzzy	Mean score
1890–1891	2,789	2,587	202	99.64
1891–1892	4,248	3,974	274	99.66
1892–1893	4,404	4,033	371	99.54
1893–1894	5,548	4,939	609	99.51
1894–1895	6,222	5,761	461	99.61
1895–1896	6,742	6,066	676	99.48
1896–1897	7,200	6,462	738	99.52
1897–1898	7,525	7,012	513	99.69
All 36 pairs combined				
All pairs	151,708	132,854	18,854	99.36